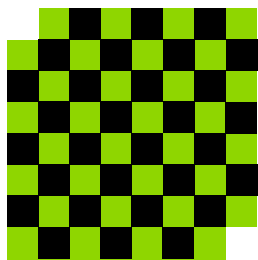


## FIND OUT MORE

### Using logic

Sometimes it's quicker to solve a puzzle by thinking than by trying it out. Here's an example. Imagine a chessboard has two opposite corners missing. Can you cover the remaining 62 squares with only 31 dominoes, each domino covering two squares? You could try to solve the puzzle by placing dominoes in different patterns, but this could take forever. With logic, you can solve it in seconds. *Hint: what colour are the two missing squares? So?*



### What's a paradox?

A paradox is a statement that seems to contradict itself when you think about it logically. Imagine you're walking towards the North Pole, with a compass showing north ahead and west on your left. If you walk across the pole and then turn around, west swaps over to the other side. It seems impossible, but it's true. Here's another. A barber in a village shaves everyone who doesn't shave themselves. Who shaves the barber? Turn to the back of the book for the answer.



# LOGIC

This branch of maths relies on *thought* rather than numbers or shapes. Given a starting point, if you can **deduce** certain things, and then link your deductions together until you have a solution, then you have solved the problem *logically*.

## Logic puzzles



### DEATH ROW

A judge is sentencing a prisoner guilty of a heinous crime. He tells the prisoner that because the crime is so bad, he will be hanged at noon within a week, but to make his suffering worse, he will not know the day of the execution until that day arrives. The prisoner thinks for a few seconds and then says, laughing, "but that means I can't be hanged". How does he know?

### THE TIGER

A mother and her son are working in the field in India. A tiger leaps out of the long grass and pins the boy to the ground with his claws. "Let him go!" cries the woman. "I will", says the tiger (it was a talking tiger) "providing you can correctly predict the fate of your child – either that I eat him or that I let him go". What should the woman predict?

The sentence on the right is true.